

A naive proposal, where it fails and what this means

Martín Soto

August 14, 2023

Motivation

We want a finite, initial belief state $\mathbb{Q}$. We will expand it to a distribution $\mathbb{P}$ over all sentences satisfying some desirable constraints (usually, we will need $\mathbb{Q}$ to already satisfy the finite versions of these constraints), and that "doesn't add new information" (or at least, information that has negative decision-theoretic consequences). We will use this $\mathbb{P}$ to $\text{argmax } \mathbb{E}(U|\mathbf{A}(\text{obs}) = a)$ when our agent $\mathbf{A}$ finds itself in different observations. (How is U defined? More on this in the **Utility** section.)

The canonical way to obtain $\mathbb{Q}$ is by running a logical inductor for some large number of rounds. To ensure $\mathbb{Q}$ satisfies the finite constraints, this inductor will be a Universal Inductor modified to satisfy the constraints in every state.

But we could also just choose any $\mathbb{Q}$ satisfying the constraints.

This is already sounding problematic, because we're using a finite belief state to guide actions in an arbitrarily complex situation, and so the usual worries of "the logical prior acting dumb" reappear. More on this in the **Reflections** section.

Step 1: Extending a Universal Inductor

It's only sensible to impose some constraints in $\mathbb{P}$ if $\mathbb{Q}$ already satisfies them, both in the sense that some of the below definitions wouldn't work, and that we just want the whole distribution to satisfy them.

We can get the below constraints by running a Universal Inductor while automatically eliminating any trades in finite worlds that are inconsistent with the constraints (that is, keeping their prices permanently at 0). The construction works as before (just in a lower dimensional simplex, and with the calculation of revenue depending on this restricted set of possible worlds). The result will still be an LI, because if a trader exploits this market, then that trader with trades in inconsistent worlds eliminated would also exploit it.

To succinctly express our constraints, consider the two following axiom schema (infinitely many axioms):

1. $\forall \bar{m} \varphi(\bar{m}) \rightarrow \varphi(\bar{i})$ for any φ and sequence of numerals $\bar{i}$.
2. $\forall m_1 \forall m_2 (\varphi_1(m_1) \wedge \varphi_2(m_2)) \leftrightarrow (\forall m_1 \varphi_1(m_1) \wedge \forall m_2 \varphi_2(m_2))$

Call it Ax. It is propositionally consistent by being a fragment of First Order Logic. We eliminate trades in worlds (finite conjunctions) that aren't consistent with Ax.

Step 2: Extrapolating to large beliefs

Although the Universal Inductors framework has infinite support by arbitrarily setting all untouched sentences to 0.5, of course we don't care about those guesses, and will only take as $\mathbb{Q}$ the worlds that have been touched by some trader (or alternatively, we could re-define Universal Inductors so that they only have this finite support). Indeed, what we're doing here can be understood as "Universal Inductors with better extrapolation".

Now fix an enumeration of all sentences so that the finite support of $\mathbb{Q}$ is an initial segment. That is, $\mathbb{Q}$ has values for all the $(\neg)\phi_1 \wedge \dots \wedge (\neg)\phi_n$ for some fixed n . We inductively define $\mathbb{P}$ by setting conditional probabilities in the enumeration order, as follows (where C is an arbitrary conjunctions, always in the order of enumeration):

- $\mathbb{P}((\neg)\phi_1 \wedge \dots \wedge (\neg)\phi_n) := \mathbb{Q}((\neg)\phi_1 \wedge \dots \wedge (\neg)\phi_n)$
- $\mathbb{P}(C \wedge \phi) := \mathbb{P}(C)$ whenever $\text{Ax} \cup \{C\} \vdash_{\text{Prop}} \phi$,
and $\mathbb{P}(C \wedge \phi) := 0$ whenever $\text{Ax} \cup \{C\} \vdash_{\text{Prop}} \neg\phi$.
- $\mathbb{P}(C \wedge \phi) := \frac{1}{2}\mathbb{P}(C)$ whenever none of the above clauses are triggered.
- Of course, $\mathbb{P}(C \wedge \neg\phi) := \mathbb{P}(C) - \mathbb{P}(C \wedge \phi)$.
And given any formula φ , we consider all finite conjunctions C with literals its prime formulas, and define $\mathbb{P}(\varphi) := \sum_{C \vdash_{\text{Prop}} \varphi} \mathbb{P}(C) = \sum_{\text{Ax} \cup \{C\} \vdash_{\text{Prop}} \varphi} \mathbb{P}(C)$.

Since by assumption $\mathbb{Q}$ was already consistent with Ax (that is, only had non-zero probability in Ax-consistent sentences), it's immediate to prove by induction that this is well-defined and that $\mathbb{P}$ is consistent with Ax.

Also, deciding whether $\text{Ax} \cup \{C\} \vdash_{\text{Prop}} \phi$ for finite C is decidable, since from a set of prime formulas Ax can only derive smaller or equal length prime formulas, with the exception of some negations of quantified sentences, from which we cannot in turn derive smaller formulas in any way. That is, we can syntactically cap the length of a possible proof.

Ax ensures $\mathbb{P}(\forall \bar{m} \phi(\bar{m})) \leq \lim_{\bar{m}} \mathbb{P}(\bigwedge_{\bar{m}} \phi(\bar{i}))$. We also get the other inequality (intuitively corresponding to "we are in a standard model"), because $\lim_{\bar{m}} \mathbb{P}(\neg \forall \bar{m} \phi(\bar{m})) \wedge \bigwedge_{\bar{m}} \phi(\bar{i}) = 0$. Indeed, the latter could only fail if a finite and Ax-consistent $C \wedge \neg \forall \bar{m} \phi(\bar{m})$ propositionally entailed infinitely many $\phi(\bar{i})$. This isn't possible with our Ax, because the only way to have this is through quantification. In some sense, the reason why this happens is that we don't propagate equality of sub-terms or equivalence of sub-formulas (more on this below).

Of course, most sentences' probabilities will still arbitrarily depend on enumeration order, and so be "off-base guesses". For example, this will be the case for any prime formula bigger than any formula in the support of $\mathbb{Q}$. We could arbitrarily fix an enumeration to make these guesses have arbitrary structure.

LUVs

We fix a syntactic procedure to, from any single-parameter formula $X(m)$, write a sentence LUV_X that (in PA) expresses " X defines a $[0, 1]$ -LUV".

We also fix a procedure to obtain a single-parameter $D_X(m)$ such that (in PA) $D_X(\underline{1})$ expresses "the $[0, 1]$ -LUV defined by X is in $[0, \frac{1}{2}]$ ", $D_X(\underline{2})$ expresses "the $[0, 1]$ -LUV defined by X is in $[0, \frac{1}{4}] \cup (\frac{1}{2}, \frac{3}{4}]$ ", etc. Before starting the above process, we rewrite $X = Y$ as $\forall m(D_X(m) \leftrightarrow D_Y(m))$. Similarly for $X > Y$ and other sentences about LUVs.

As long as what we check about LUVs are these syntactically pre-defined sentences, they will present the nice behavior we wanted related to expectations because of propositional consistency.

Other constraints

- *(This bullet point is technical, rambling and skippable)* **Propagation:** We can imagine syntactically fixing many other small properties that keep entailment decidable, similar to axiom schema 2., like for example $\forall m \varphi(m) \rightarrow \forall m(\varphi(m) \vee \phi(m))$. But we would also want something more general, completely propagating small beliefs through large beliefs, of the form $\varphi \wedge \forall \bar{m}(\phi_1 \leftrightarrow \phi_2) \rightarrow \varphi(\phi_2/\phi_1)$, where $\varphi(\phi_2/\phi_1)$ is the result of substituting the sub-formula ϕ_2 for ϕ_1 in φ (preserving the names of any parameters they take). Unfortunately, this would make entailment undecidable, by solving the word problem for groups (unless some other axioms already decide all such φ , in which case we don't care about this axiom schema). Same goes for propagating equality, as in $\varphi \wedge \forall \bar{m}(t_1(m) = t_2(m)) \rightarrow \varphi(t_2/t_1)$. (I think the problem arises by incorporating general syntactical substitution, instead of only substituting parameters, or applying one from a finite list of syntactic schemas.)

It intuitively seems like such propagation would allow for arbitrary computation.¹

Even ignoring computability, these would break the second limit inequality, because for example $\varphi(\underline{1}) \wedge \forall m(\varphi(m) \leftrightarrow \varphi(m+1)) \wedge \neg \forall m(\varphi(m))$ would be consistent, and nonetheless entail all $\varphi(\underline{i})$.

We could additionally add some induction axiom to fix this (so that when infinitely many $\varphi(\underline{i})$ are entailed, we can also entail the quantified sentence witnessing that). Would that be decidable?

Maybe we can instead find some hackish middle ground exploiting syntax, like including only $\varphi \wedge \forall \bar{m}(\phi_1 \rightarrow \phi_2) \rightarrow \varphi(\phi_2/\phi_1)$ in which ϕ_2 has quantifier depth lower than that of ϕ_1 . Not clear what this fixes.

Consider this more radical change. I've been treating numerals $\underline{m}$ as $S^m 0$, as always, so that formulas can say things like $\forall m(\varphi(m) \leftrightarrow \varphi(S(m)))$. What if instead I take every numeral to be a different atomic constant c_m (instead of a built term)? To be clear, we'll still be able to

¹This could be taken advantage of by running traders, but it's not clear how our single finite $\mathbb{Q}$ could do it (there's no single formula equivalent to "whenever you want to check the output of this computation, check the probability of the sentence with this syntactic form"). So it's not clear how bad this is on updateless grounds.

say things like $\forall m(m+1 = S(m))$, or talk about $S^m 0 := S^m c_0$. It's just, when substituting in a constant, it won't be "immediate by definition" that $c_4 = S(c_3)$. That is, instead of having $\forall m \varphi(m) \rightarrow \varphi(S^i 0)$, I'll have $\forall m \varphi(m) \rightarrow \varphi(c_i)$. And also in other situations I will check $\mathbb{P}$'s credence on $\varphi(c_i)$ instead of $\varphi(S^i 0)$, like when taking the limit. This is a way to curtail induction. "General beliefs about the relationships between numbers" won't leak into my assessments of the properties of concrete numbers. Because of that, the above breaking of the second limit inequality won't happen. This resonates with updateless worries, since oracle-checking $S^i 0$ instead already yields some additional computation, because of its syntax being an explicit tally representation. With the c_i , we could enforce "all of FOL" (although this would still be undecidable) without having to worry about the leakage of beliefs about equality. With the c_i , traders could still make hypotheses about their relationships notice their relationships (by looking at the sub-index of each constant), but our syntactic extrapolations would treat them as independent nameless constants. (This distinction might be more positive in a situation with infinite rounds of trade.) In any event, it's not clear if this change hits the right level of updatelessness.

- **Basic facts:** We can also consider enforcing some basic properties of numbers, like $\underline{m} + 1 = S\underline{m}$. But this becomes less attractive when we notice, as above, that we won't be able to propagate these equalities to use them in large sentences. Indeed, we'd need to include this as an axiom schema, as $\varphi(\underline{m} + 1) \leftrightarrow \varphi(S(\underline{m}))$ for every φ and m . This points to a general motif: we can implement more of these basic facts, but doing anything too general will break decidability. And also, we could say, suspiciously give more unwarranted ground to updatefulness. Indeed, $\underline{m} + 1 = S\underline{m}$ seems so simple or easy to compute that we "wouldn't want to be updateless about it". But that's not how updatelessness works. What differentiates decision-theoretically $\underline{m} + 1 = S\underline{m}$ from $\underline{m} \cdot \underline{n} = \underline{m} \cdot \underline{n}$. Is there some sense in which "on average, being updateful about the first makes us lose out on less"? That's already making a kind of simplicity assumption, that easier to compute things will be less prone to appear in a situation where acting updateless is better. And I don't think that's the case, especially when considering larger sentences (the larger the sentence the better it is to be updateless about it? that sounds so weird). Maybe this is one of those "simplicity properties that our reality seems to have" and so we want to hard-code it, even if it's not mathematically general? (Like finely tuning the Universal Turing Machine with which we do Solomonoff Induction.)
- **Introspection:** I would also want something like $\mathbb{P}(\ulcorner \mathbb{P}(\ulcorner \phi \urcorner) \urcorner < p) = \mathbb{1}(\mathbb{P}(\phi) < p)$. That is, when in our induction we come upon a sentence of this syntactic form containing the Gödel number for ϕ , we check what we actually believe about ϕ . As usual, apparently we get impossibility with $\phi := \ulcorner \mathbb{P}(\ulcorner \phi \urcorner) \urcorner < 1$. But actually it's more complicated: this kind of self-referential ϕ won't explicitly include its written down Gödel number. It will only contain a sub-formula ϕ' defining it. And possibly our reasoner is uncertain about which concrete numeral this sub-formula defines. What if we just require this property when the numeral $\ulcorner \phi \urcorner$ is explicitly written? (As always, this amounts to ignoring some equivalences that our reasoner might still not know.) Even in that case we could have $\mathbb{Q}_n(m = \ulcorner \phi \urcorner \leftrightarrow \phi'(m)) = 1$ (because $\mathbb{Q}$ has seen a proof. But even with this, we'd only get impossibility if we had general "propagation of equalities / sub-formulas", which for now we don't. By choosing any enumeration such that formulas containing the numeral $\ulcorner \phi \urcorner$ (and outside the support of $\mathbb{Q}$) appear after ϕ , we can just say

$$\mathbb{P}(C \wedge \ulcorner \mathbb{P}(\ulcorner \phi \urcorner) \urcorner < p) := \mathbb{P}(C) \cdot \mathbb{1}(\mathbb{P}(\phi) < p)$$

and this will be well-defined. Such an enumeration exists for any numbering with $\ulcorner \ulcorner \phi \urcorner \urcorner > \ulcorner \phi \urcorner$. But we still have a greater problem: how do we ensure $\mathbb{Q}$ itself satisfies this constraint (that is, doesn't believe dumb things either about itself, or about $\mathbb{P}$). This cannot be solved like the other cases, since the incoherence of a bet depends on the whole market. For example, if T_n wants to bet on $\varphi \wedge "Q_n(\varphi) > 0.5"$, I will want to have $Q_n("Q_n(\varphi) > 0.5") = \mathbb{1}(Q_n(\varphi) > 0.5)$. But this will constitute a discontinuity of the $\text{fix}(\mathbb{V})$ function, and so we won't be able to find a fix-point. This seems like a fundamental obstacle.

What about making it continuous but arbitrarily sharp? (Gets sharper every round) Then I think it should work. Although maybe most times it ends up at 0.5 or something.

- **Instrumentality:** We don't want to believe at the same time A and "Believing $\neg A$ is instrumentally better". In some extreme cases this can be syntactically prohibited: for example, never allowing a trade on $\varphi \wedge$ "Expected utility is highest when $Q_n(\varphi) = 0$ ". But most times the situation will be more complex, so that $\text{argmax}_p \mathbb{E}_n(U|Q_n(\varphi) = p) \in (0, 1)$. As above, intuitively, it's not clear how to treat a trader who bets on both φ and "the best $Q_n(\varphi)$ possible is p ". And as above, what's going on mathematically is that, if $\mathbb{E}_n(U|Q_n(\varphi) = p)$ is not continuous on p , $\text{fix}(\mathbb{V})$ won't be continuous (and indeed it's easy to come up with examples where there's no fix-point). Certainly in some situations the payoff is not continuous on p , and we do want $\mathbb{Q}$ to know this. Although maybe in our decision formalism, where Ω uses an LI, it is always continuous? (I don't think so, because which action we take is a binary decision, and a slight change of p can change it, and it affects U .)

As above, maybe we want to make $\mathbb{E}_n(U|Q_n(\varphi) = p)$ continuous but arbitrarily sharp. And/or make some assumptions about our decision tree so that the real payoff is continuous.

- **Averaging out:** Above I settled on the "consistency relative to Ax " because it was the True Name of what I had been trying to do from the beginning: have a finite list of syntactic rules that inform the probabilities of some sentences, but in such a way that no contradictory recommendations are ever made. For example, if whenever I have $\forall m \varphi(m)$ I want to put $\varphi(\underline{1})$ to probability 1, and whenever I have $\exists m (m < \underline{2} \wedge \neg \varphi(m))$ I want to deduce $\neg \varphi(\underline{1})$, and I have both of them in my conjunction C , then I have conflicting recommendations and the induction is not well-defined. Consistency fixes this by ensuring no such C is ever possible. But it brings the problems of undecidability for general enough rules. What if, instead, I'm comfortable naively applying such general rules, and whenever such a conflict arises I just average between the possibilities, as if the rules of extrapolation were different fixed experts (with the ability to abstain from judgement)?
- **Stability:** Long ago I had that idea about Reference Classes, in which the correctness of a Reference Class was judged by how stable its fraction of true sentences had been in the past. Instead of that, consider judging the stability of $Q_n(\varphi)$. Maybe, when turning $\mathbb{Q}$ into $\mathbb{P}$, I want to hard-code some belief about how trustworthy different $\mathbb{Q}$ opinions are, and this could be proportional to how stable their probabilities have been. Something like $\mathbb{P}("|Q_{n+m}(\varphi) - \mathbb{P}(\varphi)| \geq p") \propto \text{Average fluctuation per round in the past.}$

Utility

Wait, what do we mean by the U in $\mathbb{E}(U|\mathbf{A}(\text{obs}) = a)$?

I think we need to mean $\lim \bar{U}$, the limit of the average utility obtained per round. (And if our tree is finite, this is equivalent to total utility.)

Otherwise, in infinite trees, as in our online learning framework, we would mean $U_{>0} := \delta \cdot U_1 + \delta^2 \cdot U_2 + \dots$, the time-penalized future utility. But this seems to pose a conceptual problem: we will genuinely take future bankruptcy for a penny now.

And if instead, when evaluating the behavior of **A** in some *obs* pertaining to round n , we use $U_{>n}$, we reintroduce dynamic instability.

In online learning time-penalization made sense, since that instability in values was also coupled with the instability in learning. But in this case, where our knowledge is frozen, and we're trying to "look at everything from the outside and decide on the globally best policy", I think $\lim \bar{U}$ is the true name.

But then what if our prior thinks there's some possibility of this limit diverging? Well, those cases will completely swamp its decision-making... which is a problem with fixed-prior expected value maximization, not our approach in particular.

How can we prevent this? By having "infinity" be worth some very high number, higher than all other possible limits our prior is considering? By making the assumption that this never happens in our decision trees, and enforcing that $\mathbb{P}$ knows that with certainty?

As you mentioned, there might also be some ways to establish preferences between actions / policies / programs, even when things go infinite. But it doesn't seem like they'll be general enough. But I need to think more about this.

Reflections

I see some fundamental conceptual problems with the "ambitious updatelessness" approach. More concretely, with the idea of fixing a finite belief state, extrapolating it, and deciding actions based on its expected value calculations.

Consider the case of LUVs. We have syntactic LUV coherence with the above. But for the infinitely many large sentences, not even the simplest syntactic arbitrage of "to which things is $D_X(i)$ equivalent" will have been carried out. And I think we cannot trust our environment to always present the decision problems / logical coins in "the most convenient syntactic shape for us"². So we'll be stuck with our prior, having random credences on almost all large sentences, including ones which are "very close" to sentences we actually know. Of course, we cannot just "perform the arbitrage automatically", or "propagate the knowledge to only these very close sentences", because we don't know how to draw this line (and I would say: we care about it for its consequences, so there is no consequence-independent way to draw this line).

The above sounds especially bad in infinite / in-the-limit situations, since ultimately we will act randomly in infinitely many rounds. But maybe you were thinking of finite trees, in which in some sense the finite knowledge of our prior is sufficient and optimal.

In that case, I've got some confusions about what we're trying to get in the finite situation.

This finite tree will also need a specific way to compute logical counterfactuals (like our LI frame-

²Maybe nonetheless you want us to assume this is the case and see what happens? Well, even then we'll be stuck with infinitely many random guesses.

work). And given that, we can just argmax over the utility that a finite policy (sending decisions to actions) attains in the whole tree.

Maybe what we're looking for is some proof that "this finite program is actually optimal in the class of all programs for all decision trees in this class, and it's doing something cooler than just argmaxing", revealing some "nature of intelligence". This is the strongest possible kind of finite assurance.

Other notions of finite assurance, like "by round $10^{10^{10}}$ it doesn't have negative accumulated utility", clearly cannot work in general even with assumptions. So I guess the only things we could be saying are either

- "in the limit", as in our online learning framework
- or "comparing to other programs (possibly all programs)", as above

What I'm trying to convey is that we should be clearer on what we're trying to obtain.

I know the idea is to let the math speak, and reveal these more natural structures of decision-making. But what I have been hearing the math say lately, is that the real interesting stuff is usually in the limit, and especially far from "freezing our knowledge into a prior".

On that note, probably you were thinking of something more like "we have finite knowledge at the time we freeze it into a prior, but we're also calibrated about its fallibility and act accordingly, for example allowing our agent to consult some future information as part of its policy". And in that case, some of my worries above are indeed exaggerated. This approach is very interesting, and sounding more like your Policy Selection (and I'd say the in-the-limit framework there, where the "thinker" learns much slower than the policy but nonetheless keeps learning, is probably more natural than our "freeze the thinker at a finite time").

But what I would say here, is that I think the correct version of this will look very different from "freezing a prior". You could freeze a prior, and use that to decide on a policy that consults a more advanced LI. But this will still face the problem that in infinitely many situations the prior will have bad guesses about what policies are best. And it's not like we can "default to updatefulness" in those cases, because the prior can't distinguish them. As much as the frozen prior might have gathered some information on its fallibility, that itself is finite, and will "run out" eventually, or "not reach infinitely many places".

When freezing a prior, we are first getting some finite amount of epistemic benefit, and then freezing it and exploiting it instrumentally forever. I think in the correct version the epistemic and instrumental will be much more intertwined and in dialogue (and in particular, epistemic learning will never stop). More concretely, some of the LI fix-point ideas in **Instrumentality** and **Introspection** above seem to be getting at something natural. Although I'm not sure if anything in this space will turn out more natural than some general kind of policy selection.

The ω levels of meta-closure also feel now more relevant than ever, as in Nisan's thing, although I don't see how they could be implemented at finite times (only in-the-limit).

Ways forward

Directions close to the fixed-prior idea:

- Obtaining conceptual clarification on our objective. I don't see right now how, even if we make $\mathbb{P}$ have better properties, some of the conceptual worries will disappear (especially "doing random things in infinitely many occasions").

- Trying to develop some more technical tricks (as in **Propagation**, **Basic facts**, **Introspection** and **Instrumentality**) so that $\mathbb{P}$ has even better properties.
- Instead of that, trying to get nice things by **Averaging out** instead.
- Instead of that, trying to frame this in terms different than an induction.
- Studying how the problem with **Utility** can be resolved.
- Instead of that, going back to Sam's BLI to see if we could do something differently (maybe go back to " $\mathbb{P}$ observes $\mathbb{Q}$ from the outside").
- **Infinite bets** (see below).

Directions closer to in-the-limit intuitions:

- Trying to get instrumentally aware Universal Inductors as in **Introspection** and **Instrumentality**.
- Instead of that, looking more generally for an online process that natively intertwines epistemic and instrumental objectives.
- Trying to prove the in-the-limit optimality of Policy Selection / Vetoing Traders, and seeing which boundaries can be expanded there, or how finite assurances for them might look like.
- ω -hierarchical policy-setting, as in Nisan's thing.

Appendix: Infinite bets

When extrapolating $\mathbb{P}$ to $\mathbb{Q}$ above, some arbitrary decisions were made. One of them was setting any sentence we didn't have any information on to 0.5, instead of any other fraction. This is the more general version of "deciding how fast the probability of this sequence of conjunctions decays" (as in my previous document, with the Gaifman thing). What if we let the traders bet on this themselves, instead of setting it arbitrarily?

That is, what if a trader can provide a description of a sequence $\bar{\phi}$, and a function f , and say "in each sentence ϕ_m I want to bet $f(m)$ ".

The problem with this is some of these orders would involve moving infinite amounts of money, both when paying for the shares, and when receiving their payoffs. We might be tempted to say something like "only allow trades whose total mass (that is, $\sum_m f(m)$) is finite, or provably finite, or finite in every possible world". The problem is, these trades will be arbitrarily close to a finite trade (since the infinite tail is minuscule). And in fact, if we just allow the finite sequence of trades to be longer each day (as LIs already do because of computation), the difference goes to 0. So to get anything different, we need to deal with infinite bets natively. I want to devote some thought to this apparently crazy idea.